Algebra Exercises

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CAS MA 741

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Chapter 6

Linear Algebra

6.1 Free Modules Revisited

Exercise 6.1.1. Show $\mathbb{R} \cong \mathbb{C}$ as $\mathbb{Q}$-vector spaces.

> **Solution.** This is tricky. I think $\dim_{\mathbb{Q}} \mathbb{R} = \aleph_0$, and $\dim_{\mathbb{Q}} \mathbb{C} = 2 \dim_{\mathbb{Q}} \mathbb{R} = \dim_{\mathbb{Q}} \mathbb{R}$. Unsure how to show this. ■

Exercise 6.1.2. Consider the sets

- $\mathfrak{gl}_n(\mathbb{R}) \coloneqq \mathcal{M}_n(\mathbb{R})$;
- $\mathfrak{sl}_n(\mathbb{R}) \coloneqq \{M \in \mathfrak{gl}_n(\mathbb{R}) \mid \operatorname{tr} M = 0\}$;
- $\mathfrak{sl}_n(\mathbb{C}) \coloneqq \{M \in \mathfrak{gl}_n(\mathbb{C}) \mid \operatorname{tr} M = 0\}$;
- $\mathfrak{so}_n(\mathbb{R}) \coloneqq \left\{M \in \mathfrak{gl}_n(\mathbb{R}) \mid M + M^t = 0\right\}$;
- $\mathfrak{su}(n) \coloneqq \left\{M \in \mathfrak{gl}_n(\mathbb{C}) \mid M + M^\dagger = 0\right\}$.

Show these sets are $\mathbb{R}$-vector spaces, and compute their dimension.

> **Solution.** First, we show $\mathfrak{gl}_n(\mathbb{R})$ is a $\mathbb{R}$-vector space. We will use the componentwise matrix notation $M \coloneqq [m_{ij}]$, where m_{ij} are the components of the matrix M and range over $\{1, \ldots, n\}$. For any $r \in \mathbb{R}$, define $r[m_{ij}] = [rm_{ij}]$. Clearly, $\mathfrak{gl}_n(\mathbb{R})$ is a group, so we will simply show it is a vector space. Note that $1[m_{ij}] = [m_{ij}]$,
>
> $$(r + s)[m_{ij}] = r[m_{ij}] + s[m_{ij}],$$
>
> and
> $$(rs)[m_{ij}] = [rsm_{ij}] = r[sm_{ij}] = r\left(s[m_{ij}]\right).$$
>
> Moreover, r is an endomorphism:
>
> $$r\left([m_{ij}] + [n_{ij}]\right) = r[m_{ij} + n_{ij}] = [r\left(m_{ij} + n_{ij}\right)] = [rm_{ij} + rn_{ij}] = [rm_{ij}] + [rn_{ij}] = r[m_{ij}] + r[n_{ij}].$$
>
> Therefore, $r \mapsto r$ is a homomorphism $\mathbb{R} \to \operatorname{End}_{\mathsf{Ab}}(\mathfrak{gl}_n(\mathbb{R}))$, and $\mathfrak{gl}_n(\mathbb{R})$ is an $\mathbb{R}$-vector space. Finally, we compute the dimension. Obviously, $\mathfrak{gl}_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$. This is so much effort to prove so I'm not going to.
>
> $\mathfrak{sl}_n(\mathbb{R})$ is very difficult, but note that $\mathfrak{so}_n(\mathbb{R}) \subseteq \mathfrak{sl}_n(\mathbb{R}) \subseteq \mathfrak{gl}_n(\mathbb{R})$, so maybe that helps.

I think the basis for $\mathfrak{so}_n(\mathbb{R})$ is basically one element per item m_{ij} in the upper triangle (not counting main diagonal), put a 1 there, and put a -1 at m_{ji}. Then $M^t = -M$ for the basis set. I think this is the entire basis. ■